適用於AI部署的高效能工作範例ONTAP

NetApp Solutions

NetApp
May 09, 2022

This PDF was generated from https://docs.netapp.com/zh-tw/netapp-solutions/ai/aicp_example_high-performance_jobs_for_ontap_ai_deployments_overview.html on May 09, 2022. Always check docs.netapp.com for the latest.

目錄

適用於AI部署的高效能工作範例ONTAP 1
適用於AI部署的高效能工作範例ONTAP 1
執行單節點AI工作負載 1
執行同步分散式AI工作負載 5

適用於AI部署的高效能工作範例ONTAP

本節包括在ONTAP 將Kubernetes部署在AI Pod上時、可執行的各種高效能工作範例。

"下一步：執行單節點AI工作負載。"

適用於AI部署的高效能工作範例ONTAP

本節包括在ONTAP 將Kubernetes部署在AI Pod上時、可執行的各種高效能工作範例。

"下一步：執行單節點AI工作負載。"

執行單節點AI工作負載

若要在Kubernetes叢集中執行單節點AI和ML工作、請從部署跳接主機執行下列工作。有了Trident、您就能快速輕鬆地建立資料磁碟區、讓Kubernetes工作負載能夠存取可能含有PB資料的資料。若要從Kubernetes Pod中存取此類資料磁碟區、只需在Pod定義中指定一個PVc即可。此步驟為Kubernetes原生作業、不需要NetApp專業人員。

> 本節假設您已將您嘗試在Kubernetes叢集中執行的特定AI和ML工作負載（採用Docker容器格式）容器化。

1. 下列命令範例顯示使用ImageNet資料集的TensorFlow基準測試工作負載建立Kubernetes工作。如需ImageNet資料集的詳細資訊、請參閱 "ImageNet網站"。

   此範例工作要求八個GPU、因此可在單一GPU工作節點上執行、該工作節點具備八個或更多GPU。此範例工作可在叢集中提交、而具有八個以上GPU的工作節點不存在、或目前正與其他工作負載一起使用。如果是、則工作會維持在擱置狀態、直到該工作者節點可供使用為止。

   此外、為了將儲存頻寬最大化、包含所需訓練資料的磁碟區會在本工作所建立的Pod內掛載兩次。另外一個Volume也會掛載在Pod中。第二個磁碟區將用於儲存結果和指標。這些磁碟區會使用PVCS名稱在工作定義中參考。如需Kubernetes工作的詳細資訊、請參閱 "Kubernetes官方文件"。

   此範例所建立的Pod中、會將「medium」值為「mory」的「emtyDir"磁碟區掛載到「開發/ shm」。Docker Container執行時間所自動建立的「/dev/shm」虛擬磁碟區的預設大小、有時可能不足以滿足TensorFlow的需求。如以下範例所示、掛載「empyDir"磁碟區可提供足夠大的「/dev/shm」虛擬磁碟區。如需有關「empyDir" Volume的詳細資訊、請參閱 "Kubernetes官方文件"。

   在此範例工作定義中所指定的單一容器、其「優先」值為「true」。此值表示容器有效擁有主機的root存取權。在這種情況下會使用此註釋、因為執行的特定工作負載需要root存取權。具體而言、工作負載執行的清除快取作業需要root存取權。是否需要這種「特殊權限：真」註解、取決於您執行的特定工作負載需求。

```
$ cat << EOF > ./netapp-tensorflow-single-imagenet.yaml
apiVersion: batch/v1
kind: Job
metadata:
  name: netapp-tensorflow-single-imagenet
```

```
spec:
  backoffLimit: 5
  template:
    spec:
      volumes:
      - name: dshm
        emptyDir:
          medium: Memory
      - name: testdata-iface1
        persistentVolumeClaim:
          claimName: pb-fg-all-iface1
      - name: testdata-iface2
        persistentVolumeClaim:
          claimName: pb-fg-all-iface2
      - name: results
        persistentVolumeClaim:
          claimName: tensorflow-results
      containers:
      - name: netapp-tensorflow-py2
        image: netapp/tensorflow-py2:19.03.0
        command: ["python", "/netapp/scripts/run.py", "--
dataset_dir=/mnt/mount_0/dataset/imagenet", "--dgx_version=dgx1", "--
num_devices=8"]
        resources:
          limits:
            nvidia.com/gpu: 8
        volumeMounts:
        - mountPath: /dev/shm
          name: dshm
        - mountPath: /mnt/mount_0
          name: testdata-iface1
        - mountPath: /mnt/mount_1
          name: testdata-iface2
        - mountPath: /tmp
          name: results
        securityContext:
          privileged: true
      restartPolicy: Never
EOF
$ kubectl create -f ./netapp-tensorflow-single-imagenet.yaml
job.batch/netapp-tensorflow-single-imagenet created
$ kubectl get jobs
NAME                                        COMPLETIONS   DURATION   AGE
netapp-tensorflow-single-imagenet           0/1           24s        24s
```

2. 確認您在步驟1中建立的工作正在正確執行。下列範例命令可確認已為工作建立單一Pod（如工作定義所指定

）、而且此Pod目前正在其中一個GPU工作節點上執行。

```
$ kubectl get pods -o wide
NAME                                                READY   STATUS
RESTARTS   AGE
IP              NODE            NOMINATED NODE
netapp-tensorflow-single-imagenet-m7x92             1/1     Running     0
3m    10.233.68.61    10.61.218.154   <none>
```

3. 確認您在步驟1中建立的工作已成功完成。下列命令範例可確認工作已成功完成。

```
$ kubectl get jobs
NAME                                          COMPLETIONS   DURATION
AGE
netapp-tensorflow-single-imagenet             1/1           5m42s
10m
$ kubectl get pods
NAME                                                  READY   STATUS
RESTARTS   AGE
netapp-tensorflow-single-imagenet-m7x92               0/1     Completed
0          11m
$ kubectl logs netapp-tensorflow-single-imagenet-m7x92
[netapp-tensorflow-single-imagenet-m7x92:00008] PMIX ERROR: NO-
PERMISSIONS in file gds_dstore.c at line 702
[netapp-tensorflow-single-imagenet-m7x92:00008] PMIX ERROR: NO-
PERMISSIONS in file gds_dstore.c at line 711
Total images/sec = 6530.59125
================ Clean Cache !!! ==================
mpirun -allow-run-as-root -np 1 -H localhost:1 bash -c 'sync; echo 1 >
/proc/sys/vm/drop_caches'
==========================================
mpirun -allow-run-as-root -np 8 -H localhost:8 -bind-to none -map-by
slot -x NCCL_DEBUG=INFO -x LD_LIBRARY_PATH -x PATH python
/netapp/tensorflow/benchmarks_190205/scripts/tf_cnn_benchmarks/tf_cnn_be
nchmarks.py --model=resnet50 --batch_size=256 --device=gpu
--force_gpu_compatible=True --num_intra_threads=1 --num_inter_threads=48
--variable_update=horovod --batch_group_size=20 --num_batches=500
--nodistortions --num_gpus=1 --data_format=NCHW --use_fp16=True
--use_tf_layers=False --data_name=imagenet --use_datasets=True
--data_dir=/mnt/mount_0/dataset/imagenet
--datasets_parallel_interleave_cycle_length=10
--datasets_sloppy_parallel_interleave=False --num_mounts=2
--mount_prefix=/mnt/mount_%d --datasets_prefetch_buffer_size=2000
--datasets_use_prefetch=True --datasets_num_private_threads=4
--horovod_device=gpu >
/tmp/20190814_105450_tensorflow_horovod_rdma_resnet50_gpu_8_256_b500_ima
genet_nodistort_fp16_r10_m2_nockpt.txt 2>&1
```

4. *選用：*清除工作成品。下列命令範例顯示刪除在步驟1中建立的工作物件。

   刪除工作物件時、Kubernetes會自動刪除任何相關的Pod。

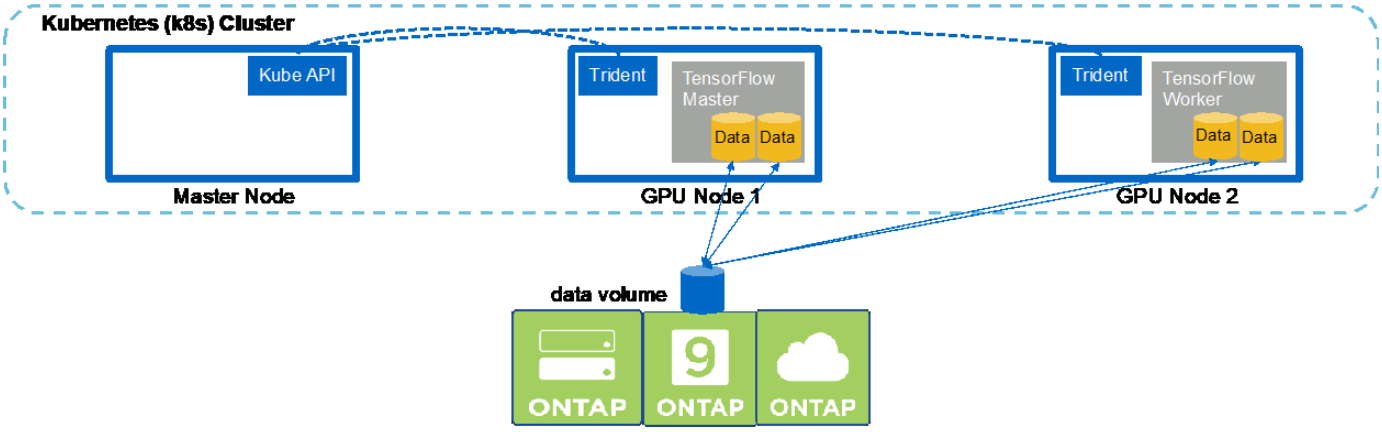
```
$ kubectl get jobs
NAME                                                COMPLETIONS   DURATION
AGE
netapp-tensorflow-single-imagenet                   1/1           5m42s
10m
$ kubectl get pods
NAME                                                       READY   STATUS
RESTARTS   AGE
netapp-tensorflow-single-imagenet-m7x92                    0/1     Completed
0          11m
$ kubectl delete job netapp-tensorflow-single-imagenet
job.batch "netapp-tensorflow-single-imagenet" deleted
$ kubectl get jobs
No resources found.
$ kubectl get pods
No resources found.
```

"下一步：執行同步分散式AI工作負載。"

執行同步分散式AI工作負載

若要在Kubernetes叢集中執行同步多節點AI和ML工作、請在部署跨接主機上執行下列工作。此程序可讓您利用儲存在NetApp磁碟區上的資料、並使用比單一工作節點更多的GPU。如需同步分散式AI工作的說明、請參閱下圖。

> 相較於非同步分散式工作、同步分散式工作有助於提升效能和訓練準確度。關於同步工作與非同步工作的優缺點的討論、不在本文的討論範圍之內。



1. 下列命令範例顯示建立一個工作者、以參與本節範例中單一節點上執行的相同TensorFlow基準測試工作之同步分散式執行 "執行單節點AI工作負載"。在此特定範例中、只會部署一名員工、因為該工作會在兩個工作節點之間執行。

   此範例的工作者部署要求八個GPU、因此可在單一GPU工作者節點上執行、該節點具備八個以上的GPU。

如果GPU工作節點的GPU功能超過八個GPU、為了發揮最大效能、您可能想要增加此數目、使其等於工作節點所使用的GPU數量。如需Kubernetes部署的詳細資訊、請參閱 "Kubernetes官方文件"。

在此範例中會建立Kubernetes部署、因為這個特定的容器化工作者永遠不會自行完成。因此、使用Kubernetes工作架構來部署IT並不合理。如果您的員工是自行設計或撰寫完成、則使用工作架構來部署您的員工可能是合理的做法。

本範例部署規格中所指定的Pod、其「hostNetwork」值為「true」。此值表示Pod使用主機工作節點的網路堆疊、而非Kubernetes通常為每個Pod建立的虛擬網路堆疊。此註釋用於此案例、因為特定工作負載仰賴Open MPI、NCCL和Horovod以同步分散的方式執行工作負載。因此、它需要存取主機網路堆疊。關於Open MPI、NCCL和Horovod的討論不在本文的討論範圍之內。是否需要此「hostNetwork：true」註釋、取決於您執行的特定工作負載需求。如需有關「hostNetwork」欄位的詳細資訊、請參閱 "Kubernetes官方文件"。

```
$ cat << EOF > ./netapp-tensorflow-multi-imagenet-worker.yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: netapp-tensorflow-multi-imagenet-worker
spec:
  replicas: 1
  selector:
    matchLabels:
      app: netapp-tensorflow-multi-imagenet-worker
  template:
    metadata:
      labels:
        app: netapp-tensorflow-multi-imagenet-worker
    spec:
      hostNetwork: true
      volumes:
      - name: dshm
        emptyDir:
          medium: Memory
      - name: testdata-iface1
        persistentVolumeClaim:
          claimName: pb-fg-all-iface1
      - name: testdata-iface2
        persistentVolumeClaim:
          claimName: pb-fg-all-iface2
      - name: results
        persistentVolumeClaim:
          claimName: tensorflow-results
      containers:
      - name: netapp-tensorflow-py2
        image: netapp/tensorflow-py2:19.03.0
        command: ["bash", "/netapp/scripts/start-slave-multi.sh",
"22122"]
```

```
        resources:
          limits:
            nvidia.com/gpu: 8
        volumeMounts:
        - mountPath: /dev/shm
          name: dshm
        - mountPath: /mnt/mount_0
          name: testdata-iface1
        - mountPath: /mnt/mount_1
          name: testdata-iface2
        - mountPath: /tmp
          name: results
        securityContext:
          privileged: true
EOF
$ kubectl create -f ./netapp-tensorflow-multi-imagenet-worker.yaml
deployment.apps/netapp-tensorflow-multi-imagenet-worker created
$ kubectl get deployments
NAME                                      DESIRED   CURRENT   UP-TO-DATE
AVAILABLE   AGE
netapp-tensorflow-multi-imagenet-worker   1         1         1
1           4s
```

2. 確認您在步驟1中建立的工作者部署已成功啟動。下列命令範例可確認已針對部署建立單一工作者Pod、如部署定義所示、而且此Pod目前正在其中一個GPU工作者節點上執行。

```
$ kubectl get pods -o wide
NAME                                                        READY
STATUS    RESTARTS   AGE
IP              NODE            NOMINATED NODE
netapp-tensorflow-multi-imagenet-worker-654fc7f486-v6725   1/1
Running   0          60s   10.61.218.154   10.61.218.154   <none>
$ kubectl logs netapp-tensorflow-multi-imagenet-worker-654fc7f486-v6725
22122
```

3. 為啟動、參與及追蹤同步多節點工作執行的主節點建立Kubernetes工作。下列命令範例可建立一個主磁片、用於啟動、參與及追蹤同一個TensorFlow基準測試工作的同步分散式執行、該工作是在本節範例的單一節點上執行 "執行單節點AI工作負載"。

   此範例主要工作要求八個GPU、因此可在具有八個以上GPU的單一GPU工作節點上執行。如果GPU工作節點的GPU功能超過八個GPU、為了發揮最大效能、您可能想要增加此數目、使其等於工作節點所使用的GPU數量。

   本範例工作定義中所指定的主Pod、其「主機網路」值為「真」、就如同在步驟1中給工作群組「主機網路」值「真」一樣。請參閱步驟1、瞭解為何需要此值的詳細資訊。

```
$ cat << EOF > ./netapp-tensorflow-multi-imagenet-master.yaml
apiVersion: batch/v1
kind: Job
metadata:
  name: netapp-tensorflow-multi-imagenet-master
spec:
  backoffLimit: 5
  template:
    spec:
      hostNetwork: true
      volumes:
      - name: dshm
        emptyDir:
          medium: Memory
      - name: testdata-iface1
        persistentVolumeClaim:
          claimName: pb-fg-all-iface1
      - name: testdata-iface2
        persistentVolumeClaim:
          claimName: pb-fg-all-iface2
      - name: results
        persistentVolumeClaim:
          claimName: tensorflow-results
      containers:
      - name: netapp-tensorflow-py2
        image: netapp/tensorflow-py2:19.03.0
        command: ["python", "/netapp/scripts/run.py", "--
dataset_dir=/mnt/mount_0/dataset/imagenet", "--port=22122", "--
num_devices=16", "--dgx_version=dgx1", "--
nodes=10.61.218.152,10.61.218.154"]
        resources:
          limits:
            nvidia.com/gpu: 8
        volumeMounts:
        - mountPath: /dev/shm
          name: dshm
        - mountPath: /mnt/mount_0
          name: testdata-iface1
        - mountPath: /mnt/mount_1
          name: testdata-iface2
        - mountPath: /tmp
          name: results
        securityContext:
          privileged: true
      restartPolicy: Never
EOF
```

```
$ kubectl create -f ./netapp-tensorflow-multi-imagenet-master.yaml
job.batch/netapp-tensorflow-multi-imagenet-master created
$ kubectl get jobs
NAME                                        COMPLETIONS   DURATION   AGE
netapp-tensorflow-multi-imagenet-master   0/1           25s        25s
```

4. 確認您在步驟3中建立的主要工作正在正確執行。下列範例命令可確認已為工作建立單一主Pod、如工作定義所示、而且此Pod目前正在其中一個GPU工作節點上執行。您也應該看到、您在步驟1中看到的工作者Pod仍在執行中、而且主要和工作者Pod正在不同的節點上執行。

```
$ kubectl get pods -o wide
NAME                                                          READY
STATUS    RESTARTS   AGE
IP              NODE            NOMINATED NODE
netapp-tensorflow-multi-imagenet-master-ppwwj               1/1
Running   0          45s   10.61.218.152   10.61.218.152   <none>
netapp-tensorflow-multi-imagenet-worker-654fc7f486-v6725   1/1
Running   0          26m   10.61.218.154   10.61.218.154   <none>
```

5. 確認您在步驟3中建立的主要工作已成功完成。下列命令範例可確認工作已成功完成。

```
$ kubectl get jobs
NAME                                        COMPLETIONS   DURATION   AGE
netapp-tensorflow-multi-imagenet-master   1/1           5m50s      9m18s
$ kubectl get pods
NAME                                                          READY
STATUS      RESTARTS   AGE
netapp-tensorflow-multi-imagenet-master-ppwwj               0/1
Completed   0          9m38s
netapp-tensorflow-multi-imagenet-worker-654fc7f486-v6725   1/1
Running     0          35m
$ kubectl logs netapp-tensorflow-multi-imagenet-master-ppwwj
[10.61.218.152:00008] WARNING: local probe returned unhandled
shell:unknown assuming bash
rm: cannot remove '/lib': Is a directory
[10.61.218.154:00033] PMIX ERROR: NO-PERMISSIONS in file gds_dstore.c at
line 702
[10.61.218.154:00033] PMIX ERROR: NO-PERMISSIONS in file gds_dstore.c at
line 711
[10.61.218.152:00008] PMIX ERROR: NO-PERMISSIONS in file gds_dstore.c at
line 702
[10.61.218.152:00008] PMIX ERROR: NO-PERMISSIONS in file gds_dstore.c at
line 711
Total images/sec = 12881.33875
================ Clean Cache !!! ==================
```

```
mpirun -allow-run-as-root -np 2 -H 10.61.218.152:1,10.61.218.154:1 -mca
pml ob1 -mca btl ^openib -mca btl_tcp_if_include enp1s0f0 -mca
plm_rsh_agent ssh -mca plm_rsh_args "-p 22122" bash -c 'sync; echo 1 >
/proc/sys/vm/drop_caches'
==========================================
mpirun -allow-run-as-root -np 16 -H 10.61.218.152:8,10.61.218.154:8
-bind-to none -map-by slot -x NCCL_DEBUG=INFO -x LD_LIBRARY_PATH -x PATH
-mca pml ob1 -mca btl ^openib -mca btl_tcp_if_include enp1s0f0 -x
NCCL_IB_HCA=mlx5 -x NCCL_NET_GDR_READ=1 -x NCCL_IB_SL=3 -x
NCCL_IB_GID_INDEX=3 -x
NCCL_SOCKET_IFNAME=enp5s0.3091,enp12s0.3092,enp132s0.3093,enp139s0.3094
-x NCCL_IB_CUDA_SUPPORT=1 -mca orte_base_help_aggregate 0 -mca
plm_rsh_agent ssh -mca plm_rsh_args "-p 22122" python
/netapp/tensorflow/benchmarks_190205/scripts/tf_cnn_benchmarks/tf_cnn_be
nchmarks.py --model=resnet50 --batch_size=256 --device=gpu
--force_gpu_compatible=True --num_intra_threads=1 --num_inter_threads=48
--variable_update=horovod --batch_group_size=20 --num_batches=500
--nodistortions --num_gpus=1 --data_format=NCHW --use_fp16=True
--use_tf_layers=False --data_name=imagenet --use_datasets=True
--data_dir=/mnt/mount_0/dataset/imagenet
--datasets_parallel_interleave_cycle_length=10
--datasets_sloppy_parallel_interleave=False --num_mounts=2
--mount_prefix=/mnt/mount_%d --datasets_prefetch_buffer_size=2000 --
datasets_use_prefetch=True --datasets_num_private_threads=4
--horovod_device=gpu >
/tmp/20190814_161609_tensorflow_horovod_rdma_resnet50_gpu_16_256_b500_im
agenet_nodistort_fp16_r10_m2_nockpt.txt 2>&1
```

6. 當您不再需要部署時、請刪除該員工部署。下列命令範例顯示刪除在步驟1中建立的工作者部署物件。

   當您刪除工作者部署物件時、Kubernetes會自動刪除任何關聯的工作者Pod。

```
$ kubectl get deployments
NAME                                        DESIRED   CURRENT   UP-TO-DATE
AVAILABLE   AGE
netapp-tensorflow-multi-imagenet-worker   1         1         1
1           43m
$ kubectl get pods
NAME                                                        READY
STATUS      RESTARTS   AGE
netapp-tensorflow-multi-imagenet-master-ppwwj              0/1
Completed   0          17m
netapp-tensorflow-multi-imagenet-worker-654fc7f486-v6725   1/1
Running     0          43m
$ kubectl delete deployment netapp-tensorflow-multi-imagenet-worker
deployment.extensions "netapp-tensorflow-multi-imagenet-worker" deleted
$ kubectl get deployments
No resources found.
$ kubectl get pods
NAME                                              READY   STATUS
RESTARTS   AGE
netapp-tensorflow-multi-imagenet-master-ppwwj   0/1     Completed   0
18m
```

7. *選用：*清除主要工作成品。下列命令範例顯示刪除在步驟3中建立的主要工作物件。

   刪除主工作物件時、Kubernetes會自動刪除任何相關的主Pod。

```
$ kubectl get jobs
NAME                                        COMPLETIONS   DURATION   AGE
netapp-tensorflow-multi-imagenet-master   1/1           5m50s      19m
$ kubectl get pods
NAME                                              READY   STATUS
RESTARTS   AGE
netapp-tensorflow-multi-imagenet-master-ppwwj   0/1     Completed   0
19m
$ kubectl delete job netapp-tensorflow-multi-imagenet-master
job.batch "netapp-tensorflow-multi-imagenet-master" deleted
$ kubectl get jobs
No resources found.
$ kubectl get pods
No resources found.
```

"下一步：效能測試。"

Copyright Information

Copyright © 2022 NetApp, Inc. All rights reserved. Printed in the U.S. No part of this document covered by copyright may be reproduced in any form or by any means-graphic, electronic, or mechanical, including photocopying, recording, taping, or storage in an electronic retrieval system-without prior written permission of the copyright owner.

Software derived from copyrighted NetApp material is subject to the following license and disclaimer:

THIS SOFTWARE IS PROVIDED BY NETAPP “AS IS” AND WITHOUT ANY EXPRESS OR IMPLIED WARRANTIES, INCLUDING, BUT NOT LIMITED TO, THE IMPLIED WARRANTIES OF MERCHANTABILITY AND FITNESS FOR A PARTICULAR PURPOSE, WHICH ARE HEREBY DISCLAIMED. IN NO EVENT SHALL NETAPP BE LIABLE FOR ANY DIRECT, INDIRECT, INCIDENTAL, SPECIAL, EXEMPLARY, OR CONSEQUENTIAL DAMAGES (INCLUDING, BUT NOT LIMITED TO, PROCUREMENT OF SUBSTITUTE GOODS OR SERVICES; LOSS OF USE, DATA, OR PROFITS; OR BUSINESS INTERRUPTION) HOWEVER CAUSED AND ON ANY THEORY OF LIABILITY, WHETHER IN CONTRACT, STRICT LIABILITY, OR TORT (INCLUDING NEGLIGENCE OR OTHERWISE) ARISING IN ANY WAY OUT OF THE USE OF THIS SOFTWARE, EVEN IF ADVISED OF THE POSSIBILITY OF SUCH DAMAGE.

NetApp reserves the right to change any products described herein at any time, and without notice. NetApp assumes no responsibility or liability arising from the use of products described herein, except as expressly agreed to in writing by NetApp. The use or purchase of this product does not convey a license under any patent rights, trademark rights, or any other intellectual property rights of NetApp.

The product described in this manual may be protected by one or more U.S. patents, foreign patents, or pending applications.

RESTRICTED RIGHTS LEGEND: Use, duplication, or disclosure by the government is subject to restrictions as set forth in subparagraph (c)(1)(ii) of the Rights in Technical Data and Computer Software clause at DFARS 252.277-7103 (October 1988) and FAR 52-227-19 (June 1987).

Trademark Information

NETAPP, the NETAPP logo, and the marks listed at http://www.netapp.com/TM are trademarks of NetApp, Inc. Other company and product names may be trademarks of their respective owners.